RouteX: Role-Based Access Control (RBAC) & Use Case Diagram Reference Guide

Project: RouteX — Logistics Delivery Operations & Truck Profitability Management System

Client / Domain Organization: Redeemed Route Logistics Inc. (RRLI)

Target Audience: Systems Analyst / UI-UX Designer / Software Engineering Groupmate

Document Purpose: Complete, authoritative specification of Actors, Roles, Permissions, Use Cases, and UML Relationships to construct an accurate and compliant **Use Case Diagram**.

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1. System Overview & Scope Boundaries

RouteX is a centralized logistics web application designed specifically for **Redeemed Route Logistics Inc. (RRLI)** to monitor completed delivery trips, audit financial revenue and expenses, manage cold chain (refrigerated) and ambient fleet vehicles, verify proof-of-delivery (POD) documentation, manage company operating expenses, and generate operational summaries.

Core Governance Tenets (Clarified Role Principles)

1. **The Administrator Can Do EVERYTHING:**

- Full system administration, complete operational execution, user management, audit logging, report template authoring, and audited record deletions.
- Freedom to record and edit deliveries from the company foundation year (`2024-01-01`) onward without the 3-day submission window restriction.
- Complete authority over all modules: Deliveries, Multi-Store Drop-offs, Fleet Trucks, Drivers & Helpers, POD Documents, Excel Staging, Company Expenses, Reports, and System Administration.

2. **Finance Can ONLY Edit Whatever Is in the Finance Page:**

- On the **Finance Page** (`/finance` -> Company Expenses), Finance holds full write authority to record new company expenses, edit existing expenses, and delete expense entries (`finance:manage`).
- In **ALL OTHER MODULES** (Deliveries, Multi-Store Drop-offs, Fleet Vehicles, Drivers/Helpers, POD Documents, User Management, and Report Template Authoring), Finance is strictly **VIEW-ONLY**.
- Finance can inspect delivery records, client drop-offs, trip revenues, and operational expenses to audit net profitability, but cannot tamper with operational dispatch or administrative configurations.

3. **The Rest Are Just View-Only, With Exceptions to Additional Admin Features:**

- **Operations Staff (`operations`):** Hands-on daily dispatch and execution coordinator. Can create and edit deliveries (subject to a strict 3-day submission window), record store drop-offs, assign vehicles and crew, upload Excel spreadsheets, attach POD receipts, and register fleet trucks. However, Operations is strictly **View-Only on the Finance Page** (cannot record or edit company expenses) and is completely **excluded from additional admin features**.
- **Auditing Staff (`hr` , `secretary` , `customer_representative`):** Purely **View-Only** across the entire application. Zero mutation authority anywhere.
- **Additional Admin Features (Strictly Exclusive to Admin):**
- User Account Management (`/user-management`): Provisioning staff, assigning roles, toggling active/inactive status, password resets, and account deletion.

- Audited Deletion Authority (`UC-35`): Permanent deletion of deliveries, trucks, company expenses, report templates, and accounts with mandatory confirmation modals.
- System Audit Logging (`/api/audit`): Chronological diff inspection of attribute state changes (old vs. new values).
- Custom Report Template Authoring (`/reports`): Creating and deleting custom reporting layouts.
- Unlimited Historical Recording: Recording operational trips prior to the 3-day operational window.

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2. Actors & Role Hierarchy

A Use Case Diagram requires clearly identified **Actors** (human or external systems interacting with the system) and their **generalization / inheritance** relationships.

2.1 The Six Canonical User Roles

RouteX implements **6 Canonical Business Roles** mapped into distinct operational profiles:

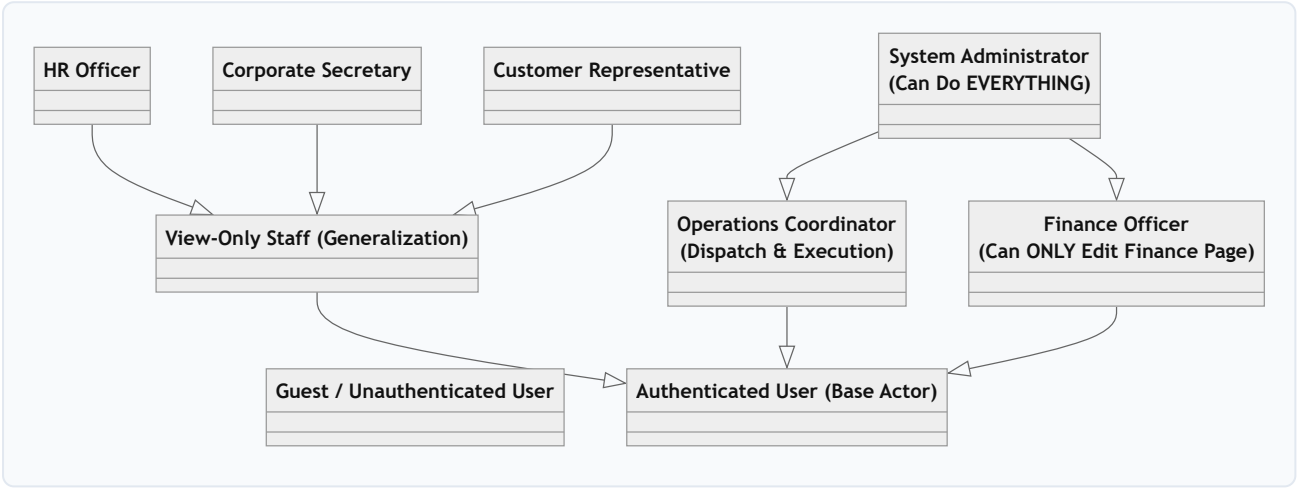
Role Name	Access Level	UI Label & Badge Tone	Primary Business Purpose & Domain Responsibil
`admin`	**Admin** (`admin`)	`Admin` *(Primary / Green)*	**Can do EVERYTHING.** Full system administration, u account management, role assignment, credential resets, audit log inspection, custom report template authoring, audited record deletions, full operational dispatch write capabilities, and company expenses management.
`finance`	**Finance Write / View-Only** (`view_only`)	`Finance` *(Neutral / Yellow)*	**Can ONLY edit whatever is on the Finance page.** F write access to the Company Expenses module (`/finance` -> record, edit, and delete company expenses). Strictly view-only across all other operatic modules (deliveries, drop-offs, trucks, drivers, POD documents, user management).
`operations`	**Operations Write Access** (`operations`)	`Operations` *(Primary / Green)*	**Daily logistics dispatch & execution.** Create and e deliveries (within 3-day window), record store drop- upload and review Excel bulk trips, attach POD files, register fleet trucks. Strictly view-only on Finance pa and restricted from additional admin features.
`hr`	**Strictly View-Only** (`view_only`)	`HR` *(Neutral)*	**Personnel auditing.** Strictly view-only access to delivery trip records, driver and helper crew assignments, daily trip wage rates, and meal allowar to prepare payroll summaries. Zero mutation author
`secretary`	**Strictly View-Only** (`view_only`)	`Secretary` *(Neutral)*	**Operational compliance.** Strictly view-only access t delivery records, POD document completeness, truc performance summaries, and operational reports. Ze mutation authority.
`customer_representative`	**Strictly View-Only** (`view_only`)	`Customer Rep` *(Neutral)*	**Customer service inquiries.** Strictly view-only acces client delivery records, store drop-off timelines, way statuses, and signed POD receipts to resolve custom claims. Zero mutation authority.

2.2 Unauthenticated & Secondary Actors

- `Guest / Unauthenticated User`: Any visitor accessing the public route (`/login`). Can submit credentials to log in, complete mandatory first-login password/security question setup, or perform password self-recovery.
- `System Audit Logger` **(Secondary / Supporting Actor)**: Automated background subsystem that intercepts administrative mutations and records immutable attribute transitions (old vs. new values).

2.3 Actor Generalization Hierarchy (UML Inheritance)

In UML Use Case Diagrams, actor generalization captures shared capabilities cleanly:



- **Inheritance & Permission Rules:**
- `Authenticated User` provides universal access to login session management (`Log Out` , `Manage Account Profile`).
- `View-Only Staff` (`HR` , `Secretary` , `Customer Rep`) inherits read-only visibility into operational screens (`Dashboard` , `Deliveries` , `POD Documents` , `Trucks` , `Reports` , `Settings`). They have **zero mutation rights**.
- `Finance Officer` inherits general viewing capabilities and adds **exclusive write authority to the Finance Page** (`Record Company Expense` , `Edit Company Expense` , `Delete Company Expense`). In all other areas, Finance acts as View-Only Staff.
- `Operations Coordinator` inherits viewing capabilities and adds write capabilities for daily operational dispatch (`Create Delivery` , `Edit Delivery` , `Store Drop-offs` , `Excel Import` , `POD Upload` , `Truck Registration`). Operations cannot edit company expenses and cannot access additional admin features.
- `System Administrator` inherits **ALL** operational and financial capabilities ("Can do EVERYTHING") and adds **Additional Admin Features** (`User Management` , `Audit Logs` , `Template Authoring` , `Audited Deletion Decisions`).

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3. Comprehensive Use Case Catalog by Subsystem

Below is the complete list of system use cases, their descriptions, primary actors, and UML relationships (`<<include>>` or `<<extend>>`).

Subsystem 1: Identity & Access Management (IAM)

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-01	**Log In**	Guest User	—	Authenticate into RouteX using username/email and RFC 9106 Argon2id password.
UC-02	**Complete First-Login Setup**	Authenticated User	`<<extend>>` UC-01 *(Condition: temporary password assigned)*	Mandatory workflow on first login: set a permanent password and register security questions.
UC-03	**Recover Password via Security Question**	Guest User	`<<extend>>` UC-01 *(Condition: user forgot password)*	Self-service password recovery by answering predefined security question.
UC-04	**Manage Account Profile**	Authenticated User	`<<include>>` UC-01	User updates own display name, username, password, or uploads profile picture avatar.
UC-05	**Log Out**	Authenticated User	`<<include>>` UC-01	Terminate the active session and invalidate HMAC-SHA256 session cookie.

Subsystem 2: Delivery Operations & Trip Management

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-06	**View Delivery Ledger**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-01	View completed delivery records in a tabular ledger with pagination and financial totals.
UC-07	**Search & Filter Deliveries**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-06	Filter deliveries by global date range, client, truck plate, driver, or POD attachment status; sort via 3-state sort controls.
UC-08	**View Delivery Details Page**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-06	Open dedicated detail page with compact record navigator, crew, vehicle specs, drop-offs, and financial breakdown.
UC-09	**Create Delivery Record**	Operations, Admin	`<<include>>` UC-01, `<<include>>` UC-10, `<<include>>` UC-11	Register a new delivery trip manually (Operations subject to 3-day window; Admin has unlimited 2024+ window).
UC-10	**Record Multi-Store Drop-off Sequence**	Operations, Admin	`<<include>>` UC-09, `<<include>>` UC-12	Specify ordered destination stores, drop-off locations, and individual delivery quantities.
UC-11	**Assign Vehicle & Crew**	Operations, Admin	`<<include>>` UC-09, `<<include>>` UC-12	Select truck plate (owned/subcontracted), assign driver, and assign up to two helpers with wage rates and meal allowances.
UC-12	**Edit Delivery Record**	Operations, Admin	`<<include>>` UC-08	Modify operational or financial attributes of an existing delivery (e.g. resolve unknown expenses).
UC-13	**Calculate Trip Financials & Margin**	System / Automated	`<<include>>` UC-09, `<<include>>` UC-12	Automatically compute revenue, total expenses, net income, and profit margin percent. Incomplete expenses mark delivery incomplete and exclude from aggregates.
UC-13B	**Delete Delivery Record**	Admin	`<<include>>` UC-08, `<<include>>` UC-35	Permanently delete an invalid delivery record via an audited confirmation screen. Exclusive to Admin.

Subsystem 3: Batch Staging & Excel Bulk Import

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-14	**Upload Excel Trip File**	Operations, Admin	`<<include>>` UC-01	Upload `.xlsx` / `.csv` spreadsheet containing multiple delivery trip rows into the staging environment.
UC-15	**Review & Validate Staged Deliveries**	Operations, Admin	`<<include>>` UC-14	Inspect parsed rows, check for missing fields, rate anomalies, or unrecognized truck plates prior to saving.
UC-16	**Confirm & Commit Batch Import**	Operations, Admin	`<<include>>` UC-15	Atomically commit validated staging rows into the active delivery database.

Subsystem 4: Proof of Delivery (POD) & Document Management

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-17	**View Proof of Delivery (POD)**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-08	View attached POD files (high-resolution PNG/JPG receipts or embedded PDF documents).
UC-18	**Download POD File**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-17	Download the original signed delivery receipt or invoice document.
UC-19	**Upload / Replace POD Document**	Operations, Admin	`<<extend>>` UC-17 *(Condition: missing or updated document)*	Attach or update a digital POD file against a delivery stop.

Subsystem 5: Fleet & Truck Profitability Analytics

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-20	**View Fleet & Profitability Grid**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-01	Review owned truck cards and subcontracted fleet vehicles with aggregated net profits and performance KPIs.
UC-21	**Filter & Rank Trucks**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-20	Filter by cargo storage capability (Refrigerated vs. Dry Cargo), wheel count, or ownership; rank by profit, revenue, or margin.
UC-22	**View Truck Operational Profile**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-20	View dedicated truck profile page containing specifications, vehicle photo, revenue history, and assigned trip log.
UC-23	**Register / Update Truck Information**	Operations, Admin	`<<include>>` UC-01	Register a new truck plate, define classification, ownership (Owned/Subcon), cargo capability, and upload vehicle image.
UC-23A	**Delete Fleet Truck**	Admin	`<<include>>` UC-22, `<<include>>` UC-35	Permanently delete a truck registration via an audited confirmation screen. Exclusive to Admin.

Subsystem 6: Company Expense Ledger (Finance Module)

> **Core Tenet:** Finance can **ONLY** edit whatever is on the Finance page (`/finance`). Admin can also edit because Admin can do **EVERYTHING**. Operations and View-Only staff can only view.

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-23B	**View Company Expense Ledger**	View-Only Staff, Operations, Finance, Admin	`<<include>>` UC-01	View company expense ledger records, total company expenses, search by description, filter by date, and sort by date or amount.
UC-23C	**Record Company Expense**	Finance, Admin	`<<include>>` UC-01	Record an overall company operating overhead expense (warehouse, rent, utilities, insurance) with positive PHP amount, date, and optional notes.
UC-23D	**Edit Company Expense**	Finance, Admin	`<<include>>` UC-23B	Update description, amount, date, or notes of an existing company expense record.
UC-23E	**Delete Company Expense**	Finance, Admin	`<<include>>` UC-23B, `<<include>>` UC-35	Permanently delete a company expense record via an audited confirmation modal.

Subsystem 7: Operational & Financial Reporting

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-24	**Generate Operational Reports**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-01	View structured reports across built-in layouts: Client Summary, Truck Summary, or Trip Ledger.
UC-25	**Filter Report by Date Range**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-24	Recompute report totals dynamically using date presets (Today, This Week, This Month, All Time) or custom dates.
UC-26	**Export Report to CSV / Excel**	Finance, Operations, Admin	`<<include>>` UC-24	Export RFC 4180-compliant CSV containing metadata, active filters, column headings, and calculated summary totals.
UC-27	**Print / Save Report as PDF**	View-Only Staff, Finance, Operations, Admin	`<<include>>` UC-24	Open clean browser print dialogue optimized for formal paper or PDF archiving.
UC-28	**Author Custom Report Template**	Admin	`<<extend>>` UC-24 *(Condition: Admin customizes layout)*	Customize column visibility, display labels, ordering, and aggregation groupings. Exclusive to Admin.
UC-29	**Delete Custom Report Template**	Admin	`<<include>>` UC-28, `<<include>>` UC-35	Delete user-created templates via an audited confirmation screen. Exclusive to Admin.

Subsystem 8: User Administration & System Auditing (Additional Admin Features)

> **Core Tenet:** All Subsystem 8 capabilities are **Additional Admin Features** exclusive to Admin. Non-admin roles (Finance, Operations, HR, Secretary, Customer Rep) have zero access.

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-30	**View User Directory**	Admin	`<<include>>` UC-01	View list of system staff, assigned roles, email addresses, and account status (`Active` vs. `Inactive`).
UC-31	**Provision New User Account**	Admin	`<<include>>` UC-30	Create account with temporary credentials and assign canonical role.
UC-32	**Modify User Role & Status**	Admin	`<<include>>` UC-30	Change staff role assignment or toggle active status (soft-disable without deleting records).
UC-32B	**Delete User Account**	Admin	`<<include>>` UC-30, `<<include>>` UC-35	Permanently delete staff user account (preventing primary admin self-deletion) via an audited confirmation screen.
UC-33	**Reset User Password (Admin-Assisted)**	Admin	`<<include>>` UC-30	Issue temporary password to unlock an account when user forgets security question answers.
UC-34	**Inspect System Audit Logs**	Admin	`<<include>>` UC-30	View audit trail capturing security events, credential updates, and record diffs showing old vs. new state changes.

Subsystem 9: Critical Confirmation & Governance

UC ID	Use Case Name	Primary Actor	Relationships & Triggers	Brief Description
UC-35	**Confirm Critical Deletion Decision**	Admin, Finance	`<<included by>>` UC-13B, UC-23A, UC-23E, UC-29, UC-32B	Present an explicit modal confirmation screen displaying entity details and destructive warnings before executing permanent deletion. Admin confirms delivery/truck/user/template deletions; Finance confirms expense deletions.

4. RBAC Permissions Matrix (CRUD & Execution Mapping)

This matrix maps system permissions directly across all roles, highlighting the three clarified principles:

Module / Permission Key	Code Identifier	Admin (Can Do EVERYTHING)	Finance (ONLY Edits Finance Pa
Dashboard			
View Dashboard & KPIs	`dashboard:view`	✅	👁 (Read)
Deliveries & Drop-offs			
View Delivery Ledger & Details	`deliveries:view`	✅	👁 (Read)
Create New Delivery	`deliveries:create`	✅ *(Unlimited)*	❌
Edit Existing Delivery	`deliveries:edit`	✅ *(Unlimited)*	❌
Delete Delivery Record	`deliveries:delete`	✅	❌
Complete / Finalize Trip	`deliveries:complete`	✅	❌
Manage Store Drop-offs	`dropoffs:manage`	✅	❌
Excel Batch Staging			
Upload Excel Trip File	`imports:upload`	✅	❌
Review & Confirm Import	`imports:review_confirm`	✅	❌
Proof of Delivery (POD)			
View & Download POD	`pod:view`	✅	👁 (Read)
Upload / Replace POD	`deliveries:edit`	✅	❌
Fleet & Profitability			
View Trucks & Profitability	`profitability:view`	✅	👁 (Read)
Register / Edit Trucks & Photos	`profitability:manage`	✅	❌
Delete Fleet Truck	`profitability:manage`	✅	❌
Company Expenses (Finance)			
View Expense Ledger & Totals	`finance:view`	✅	✅
Record & Edit Company Expenses	`finance:manage`	✅	✅ **(Write)**
Delete Company Expense Record	`finance:manage`	✅	✅ **(Delete)**
Reports			
View Report Document Previews	`reports:view`	✅	👁 (Read)
Export Reports to CSV/Excel	`reports:export`	✅	✅
Print Report Document / PDF	`reports:view`	✅	👁 (Read)
Author Custom Templates	`reports:manage_templates`	✅	❌
Delete Custom Templates	`reports:manage_templates`	✅	❌

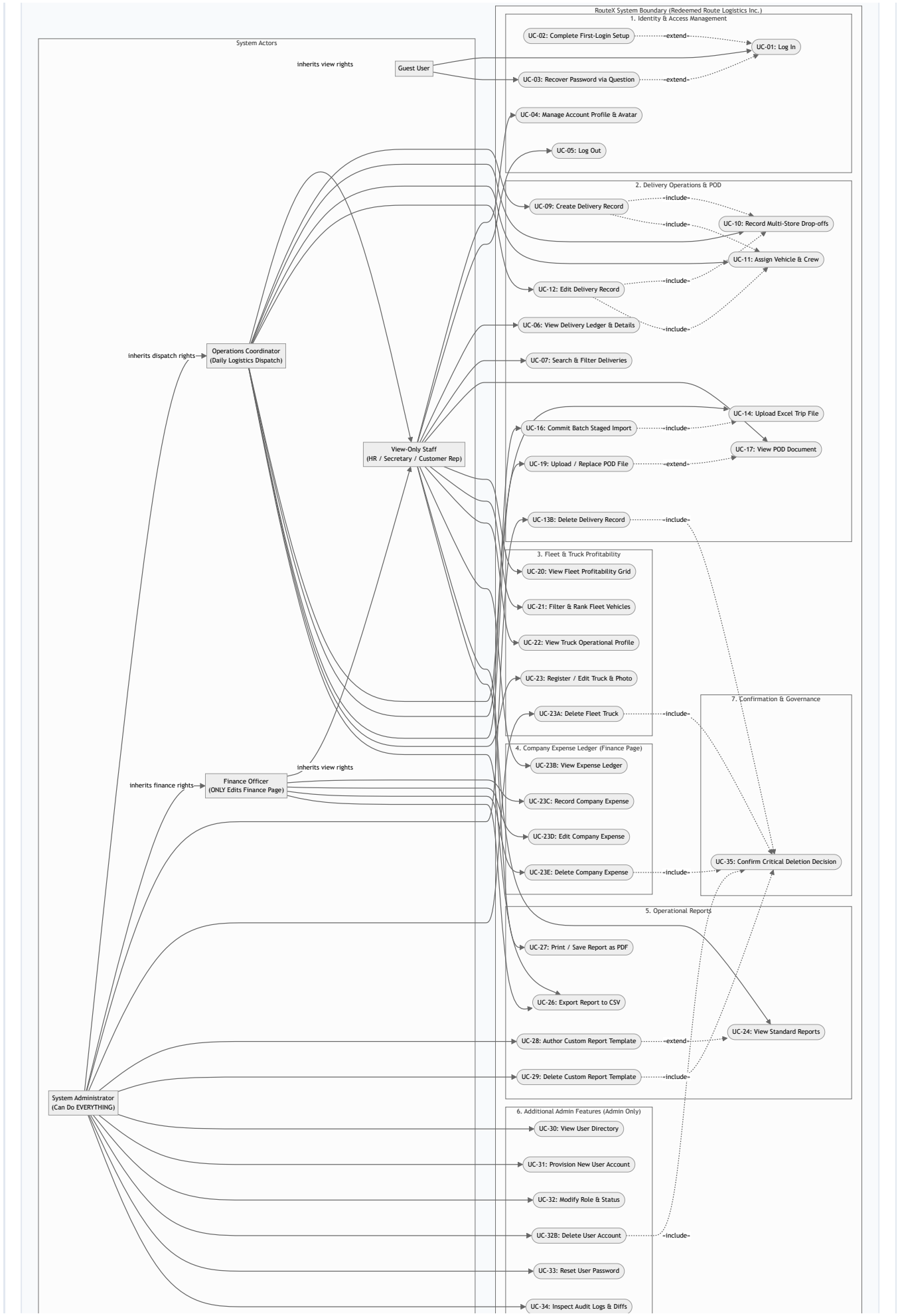
Module / Permission Key	Code Identifier	Admin (Can Do EVERYTHING)	Finance (ONLY Edits Finance Pa
Administration & IAM (Admin Only)			
View User Accounts Directory	`users:view`	✅	❌
Provision Staff Accounts & Roles	`users:manage`	✅	❌
Modify Role / Status / Resets	`users:manage`	✅	❌
Delete User Account	`users:manage`	✅	❌
Inspect System Audit Logs	`users:manage`	✅	❌
View Account Settings Page	`settings:view`	✅	👁 (Read)
Manage System Configuration	`settings:manage`	✅	❌

> **Legend:**
>
> - ✅ **Full Access:** Can view, execute, create, edit, or submit changes.
>
> - 👁 **Read Only:** Can view data, filter, search, and inspect drawers; mutation controls are hidden and server-blocked (HTTP 403 Forbidden).
>
> - ❌ **Prohibited:** Action and navigation route are completely inaccessible.

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5. Ready-to-Use Mermaid UML Diagram

Your groupmate can paste this code directly into markdown viewers, GitHub, or Mermaid Live Editor (`mermaid.live`):


System Actors
Guest User
Operations Coordinator (Daily Logistics Dispatch)
View-Only Staff (HR / Secretary / Customer Rep)
Finance Officer (ONLY Edits Finance Page)
System Administrator (Can Do EVERYTHING)
inherits view rights
inherits dispatch rights
inherits finance rights
RouteX System Boundary (Redeemed Route Logistics Inc.)
1. Identity & Access Management
UC-01: Log In
UC-02: Complete First-Login Setup
UC-03: Recover Password via Question
UC-04: Manage Account Profile & Avatar
UC-05: Log Out
2. Delivery Operations & POD
UC-06: View Delivery Ledger & Details
UC-07: Search & Filter Deliveries
UC-09: Create Delivery Record
UC-10: Record Multi-Store Drop-offs
UC-11: Assign Vehicle & Crew
UC-12: Edit Delivery Record
UC-13B: Delete Delivery Record
UC-14: Upload Excel Trip File
UC-16: Commit Batch Staged Import
UC-17: View POD Document
UC-19: Upload / Replace POD File
3. Fleet & Truck Profitability
UC-20: View Fleet Profitability Grid
UC-21: Filter & Rank Fleet Vehicles
UC-22: View Truck Operational Profile
UC-23: Register / Edit Truck & Photo
UC-23A: Delete Fleet Truck
4. Company Expense Ledger (Finance Page)
UC-23B: View Expense Ledger
UC-23C: Record Company Expense
UC-23D: Edit Company Expense
UC-23E: Delete Company Expense
7. Confirmation & Governance
UC-35: Confirm Critical Deletion Decision
5. Operational Reports
UC-24: View Standard Reports
UC-26: Export Report to CSV
UC-27: Print / Save Report as PDF
UC-28: Author Custom Report Template
UC-29: Delete Custom Report Template
6. Additional Admin Features (Admin Only)
UC-30: View User Directory
UC-31: Provision New User Account
UC-32: Modify Role & Status
UC-32B: Delete User Account
UC-33: Reset User Password
UC-34: Inspect Audit Logs & Diffs
include
extend

6. Role-Specific Business Context for Use Case Narratives

If your groupmate needs to write descriptions or narrative specifications for each role, use these verified domain summaries:

1. System Administrator (`admin`)

- **Role Clarification: The Admin can do EVERYTHING.** Full access across every module, operational capability, financial management, user management, audit logs, template authoring, and audited record deletions.
- **Why they use RouteX:** System security, account provisioning, role assignment, format configuration, accountability auditing, and ultimate operational authority.
- **Key Use Cases:** UC-30 through UC-34 (User Directory, Account Provisioning, Role Management, Credential Resets, Audit Logs with State Diffs), UC-13B/23A/23E/29/32B (Audited Deletion Decisions), UC-28 (Custom Report Template Authoring), plus all operational and finance use cases.
- **Special Rights:** Can record or edit deliveries from `2024-01-01` onward without the 3-day submission window restriction.

2. Finance Officer (`finance`)

- **Role Clarification: Finance can ONLY edit whatever is on the Finance page.** Full write authority over company operating expenses (`/finance`), but strictly view-only across all other operational modules.
- **Why they use RouteX:** To record and manage overall company overhead expenses (warehouse rent, office utilities, general administrative costs), monitor delivery revenue, and audit company net profit without altering operational trip data.
- **Key Use Cases:** UC-23B (View Expense Ledger), UC-23C (Record Company Expense), UC-23D (Edit Company Expense), UC-23E (Delete Company Expense), UC-06 (View Deliveries Ledger), UC-08 (Delivery Details Page), UC-20 (Truck Profitability Grid), UC-24 (Client & Truck Summary Reports), UC-26 (Export CSV).
- **Specific Data Watched:** Fuel expenses, highway toll receipts, helper daily allowances, client delivery rates, revenue, total company expenses, net profit (`Company Net Profit = Total Delivery Revenue - Total Delivery Expenses - Total Company Expenses`), profit margins, and incomplete expense flags.

3. Operations Coordinator (`operations`)

- **Role Clarification:** Hands-on daily dispatch execution and operational data entry. Restricted to a 3-day delivery recording window, view-only on Finance page, and completely excluded from additional admin features.
- **Why they use RouteX:** Dispatch scheduling, trip entry, store drop-off tracking, driver/helper assignments, POD image uploads, and fleet registration.
- **Key Use Cases:** UC-09 (Create Delivery), UC-12 (Edit Delivery), UC-10 (Drop-off stops), UC-11 (Assign Crew), UC-14 (Excel Bulk Upload), UC-16 (Confirm Staged Batch), UC-19 (Upload PODs), UC-23 (Register Fleet Trucks), UC-26 (Export CSV).

4. HR Officer (`hr`)

- **Role Clarification: Strictly View-Only Staff.** Zero mutation rights across the entire system.
- **Why they use RouteX:** To verify staff assignments and calculate bi-monthly crew compensation.

- **Key Use Cases:** UC-06 (Delivery Ledger), UC-08 (View Crew Details), UC-24 (Trip Ledger Report).
- **Specific Data Watched:** Driver assignments, Helper 1 and Helper 2 assignments, daily trip rates (e.g. ₱700 driver daily rate), meal allowances (e.g. ₱300), and delivery dates.

5. Corporate Secretary (`secretary`)

- **Role Clarification: Strictly View-Only Staff.** Zero mutation rights across the entire system.
- **Why they use RouteX:** To verify organizational compliance, client contract fulfillment, and document completeness.
- **Key Use Cases:** UC-06 (Deliveries), UC-17 (View POD Documents), UC-20 (Fleet Registry), UC-24 (Operational Reports).
- **Specific Data Watched:** Stamped waybill receipts, customer signature attachments on PODs, vehicle registration plate types, and delivery dates.

6. Customer Representative (`customer_representative`)

- **Role Clarification: Strictly View-Only Staff.** Zero mutation rights across the entire system.
- **Why they use RouteX:** To answer client delivery inquiries and resolve missing delivery disputes.
- **Key Use Cases:** UC-06 (Search Deliveries by Client/Waybill), UC-08 (Drop-off Sequence), UC-17 (View Signed POD).
- **Specific Data Watched:** Client name (e.g., Alfamart, SM Supermarket), waybill numbers, store drop-off timestamps, quantities delivered, and signed delivery receipts.

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7. Diagramming Best Practices & Tips for Your Groupmate

1. **Use Actor Generalization to Model the Clarified Hierarchy:**

- Draw `View-Only Staff` as the base generalization for `HR` , `Secretary` , and `Customer Rep` .
- Draw `Finance Officer` with direct association to the **Company Expense Ledger** use cases (`UC-23C` , `UC-23D` , `UC-23E`), while inheriting view rights from `View-Only Staff` .
- Draw `Operations Coordinator` with associations to operational dispatch (`UC-09` through `UC-23`), while inheriting view rights.
- Draw `System Administrator` inheriting all operations and finance capabilities and adding associations to **Additional Admin Features** (`UC-30` through `UC-34` , `UC-28` , `UC-13B` , `UC-23A`).

2. **Proper Use of `<<include>>` :**

Use `<<include>>` when a use case cannot complete without another. Example: *Creating a Delivery* always includes *Recording Drop-offs* and *Assigning Crew*. The arrow points **from** the base use case **to** the included use case: `(Create Delivery) .-> <<include>> (Assign Crew)` .

3. **Proper Use of `<<extend>>` :**

Use `<<extend>>` for optional or conditional workflows. Example: *First-Login Setup* only happens conditionally if the account is new; *Upload POD* is an extension when viewing a delivery stop that lacks a document. The arrow points **from** the extending use case **back to** the base use case: `(Complete First-Login Setup) .-> <<extend>> (Log In)` .

4. **Tool Compatibility:**

This document's structure and naming conventions can be directly copied into popular modeling tools:

- **draw.io / diagrams.net:** Standard UML Use Case stencil set.

- **Lucidchart:** UML Use Case library.
- **Visual Paradigm / StarUML:** Use Case Diagram view.
- **PlantUML:** Convert the catalog above using standard `actor` and `usecase` syntax.